

Name Method: tables for the manuscript

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In this document, I made some tables for our manuscript to illustrate our analysis process and results.

Table 1. Illustrative surnames from the Probabilistic List and the estimated probabilities of origins from each area conditional on having the corresponding surnames

Illustrative surnames	Country/Area of origin*																		
	Afghanistan	Bangladesh, India, Pakistan, Sikkim, British India	Bhutan	Brunei, Indonesia	Cambodia	Japan, Southern Ryukyu Islands	Korea, Democratic People's Republic of Korea, Republic of Korea	Laos	Malaysia	Maldives	Mongolia	Myanmar	Nepal	Peoples Republic of China, Hong Kong, Macau (Macao), Taiwan	Philippines	Singapore	Sri Lanka	Thailand	Vietnam, Democratic Republic of Vietnam, Republic of Vietnam
HUANG	0.00	0.09	0.00	0.13	0.07	0.19	0.13	0.08	0.12	0.00	0.00	0.14	0.00	96.28	0.32	0.22	0.00	0.06	1.31
CHANG	0.00	0.24	0.00	0.23	0.37	0.62	21.64	3.02	0.88	0.00	0.00	0.50	0.00	62.10	0.56	0.35	0.01	2.21	1.52
KIM	0.00	0.31	0.00	0.02	0.74	0.81	94.74	0.01	0.05	0.00	0.00	0.15	0.00	0.34	0.10	0.02	0.00	0.11	0.59
SUZUKI	0.00	0.00	0.00	0.03	0.03	95.73	0.32	0.00	0.00	0.00	0.00	0.00	0.00	0.60	0.33	0.04	0.00	0.09	0.03
NGUYEN	0.00	0.02	0.00	0.14	0.32	0.07	0.07	0.31	0.17	0.01	0.00	0.01	0.00	0.25	0.41	0.01	0.00	0.20	95.14
INTHAVONG	0.00	0.00	0.00	0.00	0.00	0.00	0.00	90.75	0.00	0.00	0.00	0.00	0.00	0.00	0.91	0.00	0.00	7.88	0.45
VICENCIO	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	1.07	55.85	0.02	0.02	0.02	0.02
SAMEER	0.00	52.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PATAN	17.74	33.87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6.45	0.00

* Note: This table exemplifies the Probabilistic List with some illustrative surnames. Each column represents one illustrative surname. Numbers in the row denote the probabilities of someone originating from the corresponding area conditional on having corresponding surnames. Probabilities in each row may not add up to 1 due to rounding error.

We also checked the number of surnames that reached 12-character limit:

[1] "N = 383, representing 0.02% of surnames."

Table 2. Numbers of surnames in the Deterministic List that are predictive for each area of origin, with the number of US-born and foreign-born SSN records having that surname

Country/Area of origin	Number of predictive surnames [*]	Number of US-born records with corresponding predictive surnames	Number of foreign-born records with corresponding predictive surnames
Afghanistan	215	12,946	18,672
Bangladesh, India, Pakistan, Sikkim, British India	6,738	1,281,803	2,506,632
Bhutan	58	4,628	22,019
Brunei, Indonesia	368	14,793	41,692
Cambodia	537	96,110	92,847
Japan, Southern Ryukyu Islands	2,995	523,987	758,857
Korea, Democratic People's Republic of Korea, Republic of Korea	277	998,728	1,289,503
Laos	574	158,842	127,920
Malaysia	119	7,105	19,192
Maldives	0	0	0
Mongolia	48	1,060	3,236
Myanmar	187	19,991	71,161
Nepal	138	12,000	63,997
Peoples Republic of China, Hong Kong, Macau (Macao), Taiwan	572	1,078,677	2,813,805
Philippines	7,794	940,978	923,734
Singapore	11	466	1,852
Sri Lanka	115	10,020	15,979
Thailand	41	1,088	2,524
Vietnam, Democratic Republic of Vietnam, Republic of Vietnam	257	594,284	1,221,744

^{*} The numbers do not add up to 22,621 because there are surnames that are equally predictive of more than one country/area.
[†] Note: The second column of this table presents the numbers of surnames that are predictive of one country/area with a 50% or higher probability. The third and fourth columns are the number of Social Security Number (SSN) application records that have the corresponding surnames.

Intext. Validation of the Probabilistic List’s ability to predict the subset of names on the Frequently Occurring Surnames from the 2010 Census that have their most frequent self-reported race/ethnicity as “Asian and Pacific Islander”

This was originally a table but we decided to only report the first row in the text and ignore the other rows as the validation done on KP cohort is more robust and complete.

There are 3170 surnames that appeared in both the Probabilistic List and the list of Frequently Occurring Surnames from the 2010 Census. The sensitivity and specificity of our list are 87.02 and 97.99, respectively. And the PPV of our list predicting the surnames in the list of Frequently Occurring Surnames from the 2010 Census is 77.41.

Table 3. Validation of the Deterministic List’s ability to predict foreign-born population’s ethnicity through cross validation

Country/Area of origin	PPV (%)	Sensitivity (%)	Specificity (%)
Afghanistan	75.95	51.44	99.96
Bangladesh, India, Pakistan, Sikkim, British India	87.40	98.84	95.94
Bhutan	62.42	47.71	99.92
Brunei, Indonesia	82.72	68.26	99.93
Cambodia	75.96	62.79	99.78
Japan, Southern Ryukyu Islands	94.56	96.65	99.55
Korea, Democratic People’s Republic of Korea, Republic of Korea	80.52	93.51	97.17
Laos	71.86	71.86	99.63
Malaysia	73.01	22.44	99.95
Maldives	33.33	1.16	100.00
Mongolia	97.00	93.01	100.00
Myanmar	72.89	66.73	99.81
Nepal	72.61	72.41	99.82
Peoples Republic of China, Hong Kong, Macau (Macao), Taiwan	87.96	93.62	95.39
Philippines	86.74	95.03	98.67
Singapore	65.65	5.18	99.99
Sri Lanka	79.63	75.10	99.97
Thailand	86.64	2.84	100.00
Vietnam, Democratic Republic of Vietnam, Republic of Vietnam	91.49	92.67	98.82

Note: The table presents the result of validating the Deterministic List through four-fold cross-validation. In each iteration, 75% of the sample is used to estimate a Probabilistic List. The country/area that each surname predicts with a probability of 50% or higher is then compared with the country/area of birth of samples in the set-aside 25% test set. The results are summarized into a confusion matrix, and Positive Predictive Value (PPV), sensitivity, and specificity are calculated. The table shows the mean results from the four test-set evaluations.

Table 4. Sensitivity and specificity of the Probabilistic List when used to predict Asian-origin ethnicity within the Kaiser Permanente Northern California (KPNC) cohort

Country/Area of origin	Number of populations in KPNC self-reported Asian Cohort				Sensitivity (%)				Specificity (%)			
	Total	<=39	40-59	>=60	Total	<=39	40-59	>=60	Total	<=39	40-59	>=60
Asia	1,085,429	599,797	334,721	150,911	71.51	71.81	71.14	71.14	97.32	97.07	97.37	97.88
Afghanistan	1,458	1,002	368	88	22.02	22.55	22.28	14.77	99.95	99.94	99.96	99.98
Bangladesh, India, Pakistan, Sikkim, British India	110,563	70,824	28,601	11,138	69.93	67.76	72.60	76.89	99.27	99.19	99.29	99.45
Indonesia	1,928	938	673	317	20.49	20.68	21.99	16.72	99.99	99.99	99.99	99.99
Cambodia	5,425	3,488	1,553	384	42.18	41.86	43.66	39.06	99.94	99.94	99.95	99.95
Japan	24,844	6,331	8,540	9,973	58.45	53.07	57.26	62.89	99.80	99.79	99.81	99.79
Korea, Democratic People's Republic of Korea, Republic of Korea	20,492	7,918	7,587	4,987	78.49	76.61	79.90	79.35	99.40	99.50	99.37	99.22
Laos	5,064	2,566	2,025	473	38.19	34.53	42.02	41.65	99.94	99.91	99.96	99.98
Malaysia	702	315	285	102	11.40	12.70	9.47	12.75	99.95	99.96	99.94	99.93
Myanmar	1,893	786	845	262	45.85	49.49	47.34	30.15	99.97	99.97	99.97	99.98
Nepal	2,310	1,683	558	69	59.91	60.78	59.14	44.93	99.97	99.97	99.97	99.99
Peoples Republic of China, Hong Kong, Macau (Macao), Taiwan	206,370	83,837	76,500	46,033	76.53	75.80	78.41	74.74	99.43	99.43	99.39	99.51
Philippines	192,565	84,867	68,394	39,304	38.93	38.58	38.40	40.63	99.26	99.14	99.33	99.47
Singapore	245	107	104	34	4.08	4.67	2.88	5.88	99.99	99.99	99.99	99.99
Sri Lanka	562	268	204	90	25.09	21.64	29.41	25.56	99.99	99.99	99.99	99.99
Thailand	3,846	2,268	1,084	494	0.52	0.35	0.55	1.21	100.00	100.00	100.00	100.00
Vietnam, Democratic Republic of Vietnam, Republic of Vietnam	56,561	25,667	22,744	8,150	85.97	84.75	87.26	86.23	99.64	99.60	99.64	99.75

Note: The table presents the result of validating the Probabilistic List within the KPNC cohort. The Probabilistic List is estimated with the Social Security Number application record through 2013 using Expectation-Maximization method. KPNC is a large, integrated health care delivery system where >4.6 million members receive comprehensive inpatient, emergency and outpatient care through 21 hospitals and >260 clinics across Northern California. The Probabilistic List's most probable origin for each surname were compared with the self-reported ethnicity of individuals carrying the same surname in the cohort. Sensitivity, specificity, and PPV in both the self-reported Asian race group and in all known Asian-origin ethnicity groups in the KPNC cohort were calculated for the whole cohort and by age group. Multiethnic Asian and Unknown Asian categories were included only to validate the performance in identifying the Asian race, not specific ethnicities. KPNC did not have an ethnicity choice for Brunei, thus the Indonesia origin here is listed independently. KPNC also did not have ethnicity choices for Maldives and Mongolia, thus these two origins were not evaluated.

Table 5. Positive predictive value (PPV) of the Probabilistic List when used to predict Asian-origin ethnicity within the Kaiser Permanente Northern California (KPNC) cohort, calibrated by geography

Country/Area of origin	Nationwide												San Francisco metropolitan area											
	Prevalence (%)				PPV (%)				PPV conditional on Asian (%)				Prevalence (%)				PPV (%)				PPV conditional on Asian (%)			
	Total	<=39	40-59	>=60	Total	<=39	40-59	>=60	Total	<=39	40-59	>=60	Total	<=39	40-59	>=60	Total	<=39	40-59	>=60	Total	<=39	40-59	>=60
Asia	5.77	5.91	6.43	5.77	62.05	60.63	65.00	62.05	NA	NA	NA	NA	27.27	26.28	28.78	27.64	90.91	89.73	91.62	92.76	NA	NA	NA	NA
Bangladesh, India, Pakistan, Sikkim, British India	1.49	1.77	1.56	1.49	59.23	60.09	61.79	59.23	98.25	98.17	98.50	98.25	5.43	6.87	5.37	2.28	84.62	86.06	85.31	76.50	97.43	97.64	97.84	95.39
Cambodia	0.08	0.09	0.08	0.08	36.65	39.20	42.41	36.65	74.92	73.05	77.33	74.92	0.18	0.21	0.15	0.16	56.12	59.15	57.44	55.24	57.10	56.39	57.29	57.29
Indonesia	0.03	0.02	0.04	0.03	34.27	33.19	44.28	34.27	61.95	60.75	72.46	61.95	0.10	0.10	0.13	0.08	67.67	66.47	74.40	57.14	57.05	56.96	67.51	49.59
Japan	0.23	0.14	0.28	0.23	40.27	25.69	45.77	40.27	90.98	84.18	91.39	90.98	0.84	0.45	1.01	1.54	71.32	53.08	75.41	82.37	88.15	78.72	89.13	93.32
Korea, Democratic People's Republic of Korea, Republic of Korea	0.45	0.41	0.55	0.45	37.15	38.62	41.27	37.15	71.04	72.12	73.80	71.04	1.11	1.10	1.31	0.88	59.45	63.09	62.69	47.34	54.06	59.14	57.99	37.97
Laos	0.05	0.05	0.07	0.05	25.85	16.68	42.52	25.85	55.97	40.85	72.15	55.97	0.15	0.16	0.17	0.11	48.59	37.34	63.89	69.41	41.09	30.52	57.14	65.06
Malaysia	0.01	0.01	0.01	0.01	1.49	2.27	1.22	1.49	4.51	7.01	3.80	4.51	0.02	0.02	0.03	0.02	5.30	6.36	4.67	4.14	3.42	4.51	3.27	2.36
Myanmar	0.06	0.08	0.05	0.06	47.80	57.23	46.20	47.80	79.94	83.48	79.54	79.94	0.12	0.11	0.17	0.09	65.38	64.09	73.44	58.88	62.41	58.86	72.97	46.83
Nepal	0.06	0.09	0.05	0.06	54.88	63.30	49.40	54.88	86.22	87.47	81.54	86.22	0.14	0.19	0.12	0.06	74.05	79.52	71.02	71.70	74.79	76.97	70.50	70.02
Peoples Republic of China, Hong Kong, Macau (Macao), Taiwan	1.35	1.33	1.49	1.35	64.82	64.22	65.98	64.82	91.18	91.10	90.29	91.18	11.23	9.75	11.60	14.08	94.44	93.49	94.40	96.15	95.79	95.20	95.29	97.25
Philippines	0.89	0.77	1.10	0.89	32.19	25.89	39.01	32.19	93.30	91.27	94.67	93.30	5.38	4.56	6.01	6.46	74.95	68.18	78.57	84.10	94.70	93.27	95.62	96.13
Sri Lanka	0.02	0.02	0.02	0.02	30.53	28.18	36.97	30.53	73.27	70.95	76.35	73.27	0.04	0.04	0.07	0.01	51.02	45.93	66.69	26.58	56.89	53.10	70.44	30.77
Thailand	0.06	0.05	0.08	0.06	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	0.13	0.11	0.20	0.10	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Vietnam, Democratic Republic of Vietnam, Republic of Vietnam	0.57	0.54	0.69	0.57	57.80	53.71	62.87	57.80	81.58	78.36	83.70	81.58	1.45	1.37	1.70	1.31	77.84	74.68	80.77	82.10	68.43	65.06	72.02	71.79

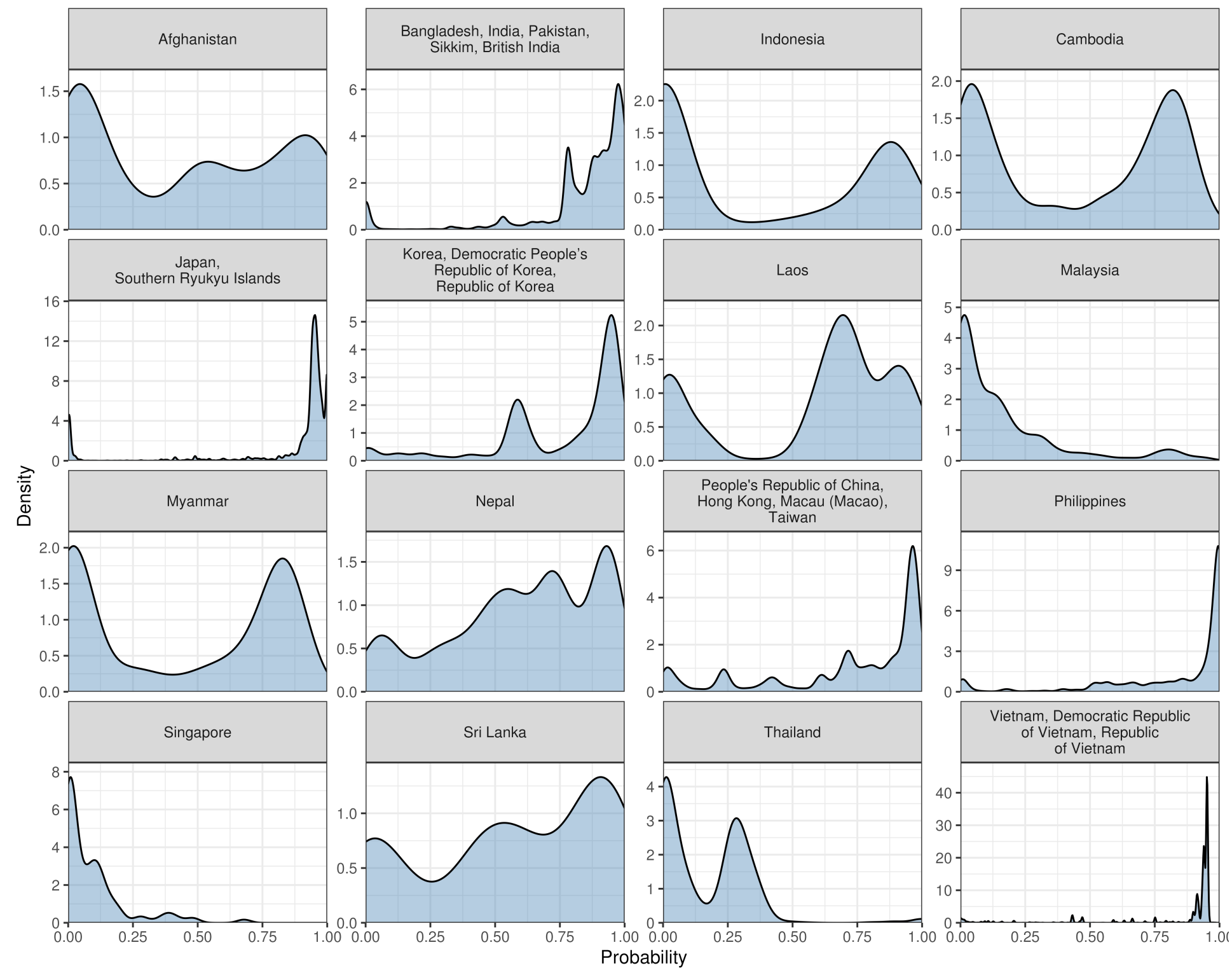
Note: The table presents the result of validating the Probabilistic List within the KPNC cohort. The Probabilistic List is estimated with the Social Security Number application record through 2013 using Expectation-Maximization method. KPNC is a large, integrated health care delivery system where >4.6 million members receive comprehensive inpatient, emergency and outpatient care through 21 hospitals and >260 clinics across Northern California. The Probabilistic List's most probable origin for each surname were compared with the self-reported ethnicity of individuals carrying the same surname in the cohort. Positive predictive value (PPV) in both the self-reported Asian race group and in all known Asian-origin ethnicity groups in the KP cohort were calculated. We then used the prevalence of Asian race group in the U.S. and San Francisco metropolitan area from the 2018-2022 American Community Survey 5-Year Estimates (ACS5) to estimate the PPVs in the corresponding area. Multiethnic Asian and Unknown Asian categories were included only to validate the performance in identifying the Asian race, not specific ethnicities. KPNC did not have an ethnicity choice for Brunei, thus the Indonesia origin here is listed independently. KPNC also did not have ethnicity choices for Maldives and Mongolia, thus these two origins were not evaluated. ACS5 did not include Afghans and Singaporean as ethnicity choices, thus these two origins were not evaluated either. We used estimates from ACS5's San Francisco-Oakland-Hayward metropolitan area estimates, which is labeled in the MET2013 variable.

Table 6. Population distribution estimates using the Probabilistic List compared with observed frequencies in the Kaiser Permanente Northern California (KPNC) cohort

Country/Area of origin	Observed frequency in KPNC (%)	Predicted probabilities (%)	Absolute difference	Symmetric relative error (%)
Afghanistan	0.23	0.17	0.06	30.00
Indonesia	0.30	0.38	0.08	23.53
Myanmar	0.30	0.71	0.41	81.19
Cambodia	0.85	1.20	0.35	34.15
Sri Lanka	0.09	0.10	0.01	10.53
People's Republic of China, Hong Kong, Macau (Macao), Taiwan	32.51	32.74	0.23	0.70
Japan, Southern Ryukyu Islands	3.91	3.57	0.34	9.09
Laos	0.80	0.98	0.18	20.22
Malaysia	0.11	1.15	1.04	165.08
Nepal	0.36	0.50	0.14	32.56
Philippines	30.33	15.86	14.47	62.65
Bangladesh, India, Pakistan, Sikkim, British India	17.42	15.67	1.75	10.58
Singapore	0.04	0.53	0.49	171.93
Thailand	0.61	0.61	0.00	0.00
Vietnam, Democratic Republic of Vietnam, Republic of Vietnam	8.91	13.58	4.67	41.53
Korea, Democratic People's Republic of Korea, Republic of Korea	3.23	6.23	3.00	63.42
Weighted mean absolute error	NA	NA	5.30	NA

Note: The table presents the result of validating the Probabilistic List at the population level against the observed frequencies in the KPNC cohort. The Probabilistic List is estimated with the Social Security Number application record through 2013 using Expectation-Maximization method. KPNC is a large, integrated health care delivery system where >4.6 million members receive comprehensive inpatient, emergency and outpatient care through 21 hospitals and >260 clinics across Northern California. For each individual, the predicted probability of each origin was calculated based on their surname, and these probabilities were averaged by origin across the entire population to estimate group-level distributions. Absolute difference = |Predicted % - Observed %|. Symmetric relative error = |Predicted % - Observed %| / ((Predicted + Observed)/2) × 100%. Weighted mean absolute error weights each group's error by its observed frequency in the population.

Figure 1. Distribution of predicted probabilities for each self-reported Asian-origin ethnicity using the Probabilistic List in the Kaiser Permanente Northern California (KPNC) cohort



Sample Table. Numbers of foreign-born records from corresponding country/area - Leftout

Country/Area of birth	Number of foreign-born records
Afghanistan	32,139
Bangladesh, India, Pakistan, Sikkim, British India	2,273,636
Bhutan	33,841
Brunei, Indonesia	58,115
Cambodia	136,556
Japan, Southern Ryukyu Islands	749,436
Korea, Democratic People's Republic of Korea, Republic of Korea	1,164,951
Laos	132,118
Malaysia	105,638
Maldives	150
Mongolia	3,502
Myanmar	98,144
Nepal	73,658
Peoples Republic of China, Hong Kong, Macau (Macao), Taiwan	2,768,925
Philippines	890,207
Singapore	46,230
Sri Lanka	18,651
Thailand	85,474
Vietnam, Democratic Republic of Vietnam, Republic of Vietnam	1,263,853
All the other countries and regions	668,407